

HAMZA SARWAR

Software Engineer | Full Stack MERN Developer | AI/ML Enthusiast

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PROFESSIONAL SUMMARY

BS Computer Science graduate and Full Stack MERN Developer with hands-on experience building scalable web applications, RESTful APIs, and secure authentication systems, plus foundational AI/ML knowledge from NAVTTC training. Currently a Software Developer Trainee at the Qalam Training Program, where I contributed to Zarkhaiz Pakistan, a full-stack agriculture marketplace, and independently built StoryMe-AI, an AI-powered storybook platform using Google Gemini. Skilled in React.js, Node.js, Express.js, MongoDB, and PostgreSQL, with a strong interest in AI-enabled solutions to real-world problems. Seeking a Full Stack role to build innovative, scalable applications while growing my software engineering and AI expertise.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), Python

Frontend: HTML5, CSS3, React.js, Tailwind CSS, Vite

Backend: Node.js, Express.js, REST APIs, JWT Authentication, Zod, Multer, Nodemailer, Bcrypt, MongoDB, PostgreSQL

AI / Machine Learning: NumPy, Pandas, Seaborn, Scikit-learn, TensorFlow, ML Model, Google Gemini API, OpenAI API

Tools & Platforms: Git, GitHub, GitLens, Postman, VS Code, Streamlit, Vercel

EXPERIENCE

Software Developer Trainee (MERN Stack) — Qalam Training Program

Feb 2026 – Present

- Developed reusable and responsive React.js components, improving UI and maintainability across the application.
- Designed and implemented RESTful APIs using Node.js and Express.js, following MVC architecture.
- Built and delivered full-stack production-style projects during the training program: Zarkhaiz Pakistan, a full-stack agriculture marketplace, and StoryMe-AI, an AI-powered personalized storybook generation platform.
- Collaborated with team members using Git and GitHub across multiple branches, debugging and resolving frontend and backend issues to improve application stability and user experience.

PROJECTS

Zarkhaiz Pakistan – Full-Stack Agriculture Marketplace (MERN Stack)

[project repo link](#)

- Designed and implemented secure authentication with JWT, OTP verification, and role-based authorization for users, sellers, and admins.
- Built a feature-based backend (routes, controllers, services, repositories) covering marketplace, cart, wishlist, orders, reviews, seller onboarding, and payouts, with real-time chat via Socket.IO and Cloudinary image uploads, connected to a responsive React.js and Tailwind CSS frontend.

StoryMe-AI – AI-Powered Personalized Storybook Platform

[project repo link](#)

- Built a full-stack platform that generates personalized children's storybooks by transforming a child's photo and prompt into AI-generated illustrated story pages using the Google Gemini API.
- Designed a feature-based REST API using Node and PostgreSQL (Drizzle ORM) with authentication (Better Auth), automated PDF/PowerPoint export, and 100+ automated backend tests, paired with a React.js/Vite/Tailwind frontend with role-based dashboards for user, learning, and admin experiences.

Custom AI Agent with API Integration

[project repo link](#)

- Developed a custom AI agent capable of processing user queries and generating responses using the OpenAI API, Tested agent functionality using predefined data to ensure accurate and consistent responses.

EDUCATION

Bachelor of Science in Computer Science (BSCS) — University of the Punjab, Lahore

2022 – 2026

CERTIFICATIONS

- Artificial Intelligence & Machine Learning — NAVTTC | Introduction to Programming Using JavaScript — Cisco Networking Academy